# Minsik Choi

*2.33, level 2, Hanna Neumann Building, Australian National University*

+61439179994 | minsik.choi@anu.edu.au | yorkcla.github.io | linkedin.com/in/yorkcla

## Profile

**Sound, Music, and Interaction: How can music shape everyday sonic interactions through computational tools and AI?**

I investigate everyday interactions and creative practices, develop AI-assisted design models and support tools across expertise levels, and evaluate their impact through mixed-methods user studies. My research interests include how musical principles can inform everyday interactions while everyday design practices, in turn, enrich musical activities, and what enables creativity in AI-mediated interactions.

*Human–Computer Interaction, Sonic Interaction Design, Sound and Music, Auditory Interface, Creative Computing, Creativity Support Tools*

## Education

### Australian National University

Ph.D. Computer Science (*Human-Computer Interaction*) *Mar 2023 - Current*

- Tonal Tension and Release in Sonic Interaction Design: A Framework and Creativity Support Tools
- Sound, Music, and Creative Computing Lab, Supervised by Dr. Charles Patrick Martin

### Seoul National University

M.S. Industrial Engineering (*Human Factors*) *Mar 2021 - Feb 2023*

- Validation and Preference User Test of In-Vehicle Virtual Sound
- Center for Human Interface Systems, Supervised by Dr. Myung Hwan Yun

### Seoul National University

B.M. Musicology and Music Theory *Mar 2013 - Feb 2020*

- Historical Interpretation of Clarinet Repertoire and Ethnomusicology, Cum Laude, Supervised by Dr. Eungi Min

### Sunhwa Arts High School

Clarinet Performance *Mar 2010 - Feb 2013*

- Student President and Academic Excellence Awards

## Publications

Musical Applications in Interactive Sonification: A Scoping Review

Minsik Choi, Charles Patrick Martin

*Proceedings of the 31st International Conference on Auditory Display*, 2026, Barcelona, Spain

A Musical Tension and Release Framework for Sonic Interaction Designers

Minsik Choi, Josh Andres, Alexander Hunter, Charles Patrick Martin

*Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, 2025, Yokohama, Japan

GenAICHI 2025: Generative AI and HCI at CHI 2025

Michael Muller, Lydia B Chilton, Mary Lou Maher, Charles Patrick Martin, Minsik Choi, Greg Walsh, Anna Kantosalo

*Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, 2025, Yokohama, Japan

Scenario-Based Design to Envision How GenAI Can Support Sound Design Practices

Minsik Choi, Josh Andres, Alexander Hunter, Charles Patrick Martin

*Proceedings of the 36th Australian Computer-Human Interaction Conference* (*OzCHI*), 2024, Brisbane, Australia

Tonal Cognition in Sonification: Exploring the Needs of Practitioners in Sonic Interaction Design

Minsik Choi, Josh Andres, Charles Patrick Martin

*Proceedings of the 19th International Audio Mostly Conference* (*AM*), 2024, Milan, Italy

Subjective Perception based on Acoustical Parameters for In-Vehicle Virtual Sound

Minsik Choi, Woochul Jung, Sungho Kim, May Jorella Lazaro, Jeonghwan Lee, Soyoun Moon, Myung Hwan Yun

*AES International Conference on Automotive Audio*, 2024, Göteborg, Sweden

Music Theoretical Approach to Auditory Interface Design: Progressive, Explainable, and Accessible

Minsik Choi

*Proceedings of the 15th ACM Conference on Creativity and Cognition* (*C&C*), 2023

Factors Affecting Interior Acceleration Sound Preference for Electric Vehicles: a Path Analysis

May Jorella Lazaro, Sungho Kim, Adriance Wilfred Anggang, Minsik Choi, Myung Hwan Yun

*Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 2022

Design and Evaluation of Electric Vehicle Sound Using Granular Synthesis

May Jorella Lazaro, Sungho Kim, Minsik Choi, Kichang Kim, Dongchul Park, Soyoun Moon, Myung Hwan Yun

*The Journal of the Audio Engineering Society*. Audio Engineering Society, 2022

Human-Centered Design in Acoustics Education for Undergraduate Music Majors

Minsik Choi, Max Kapur

*The Journal of the Acoustical Society of America*. Acoustical Society of America, 2022

# Experience

### Industry Research Intern

Dolby Laboratories, Inc. *May 2026 - Sep 2026*

- Audiovisual multimodal interaction design model based on tension and release for in-vehicle spatial audio

### PhD Student Researcher

Sound, Music, and Creative Computing Lab *Mar 2023 - Current*

- Sonic interaction design tool development combining user workflows and tonal structures with AI assistance
- Prototyping using JavaScript, React, p5.js, Tone.js, Tonal.js, Python, GenAI APIs, and Figma
- User studies using interviews, focus groups, scenarios, lab experiments, and thematic/statistical analysis
- PRISMA-based scoping review and design framework development as foundational work

### Master's Student Researcher

Center for Human Interface Systems *Mar 2021 - Feb 2023*

- User evaluation on in-vehicle virtual sound with Harman International; an in-vehicle experiment with 30 musicians
- Industry projects involving user experiment design and execution with descriptive and inferential statistical analysis

### Tutor (course teaching) and Teaching Assistant

Australian National University and Seoul National University *Sep 2019 - Dec 2025*

- Human–Computer Interaction, undergraduate and graduate, *COMP3900/6390 tutor ANU 2023, 2024, 2025*
- Sound and Music Computing, undergraduate and graduate, *COMP4350/8350 tutor ANU 2025*
- Art and Interaction Computing, undergraduate and graduate, *COMP1720/6720 tutor ANU 2023*
- Human Factors, undergraduate, *406.304 TA SNU 2021*
- Human Interface Design, undergraduate, *406.427A TA SNU 2021*
- History of Western Classical Music, undergraduate, *650.2242 TA SNU 2019*

### Student Mentoring

Australian National University *Aug 2025 - Jun 2026*

- Designing Soundscapes for Verbal Interactions, Honours student, *Anneysha Sarkar*
- Moderating Respiration Rate through Musical Parameter Control, Honours student, *Aditya Grama*

### Professional Service

Conferences in HCI, Sound, and Music *Mar 2023 - Current*

- Workshops Co-Chair (NIME '25); Workshop Co-Organiser (GenAICHI '25); Regional Chapter Leader (KASEA ANU '25)
- Reviewer (CHI '25–'27; DIS '25; NIME '25; OzCHI '25–'26; *The Journal of the Acoustical Society of America*)
- Program Committee (OzCHI '26 Long Papers)

### Music Education Project Manager

SNU, Naver, and Korea Educational Broadcasting System — *Jan 2021 - Jun 2022*

- Planned and coordinated lesson programmes with SNU music faculty
- Led on-site filming and production for large-scale online music education
- Delivered 300 online music lessons published on Naver

### Start-up Experiences

Ailys and Outliers — *Apr 2020 - Aug 2020*

- Ailys: Developed Middle East market-entry strategies and led Southeast Asia channel acquisition for an ML solution
- Outliers: Designed BTL-focused user events as part of the early-stage setup of a social networking service

## Projects

### PhD Project

Sound, Music, and Creative Computing Lab — *Mar 2023 - Current*

- Tool refinement and evaluation; CadenceFlow: A Design Support Tool for Creating Sonic Interactions through Tonal Tension and Release
- Framework design with GenAI; A Musical Tension and Release Framework for Sonic Interaction Designers
- Prototyping with GenAI; Scenario-Based Design to Envision How GenAI Can Support Sound Design Practices
- Needs finding user study; Tonal Cognition in Sonification: Exploring the Needs of Practitioners in Sonic Interaction Design
- Systematic scoping review; Musical Applications in Interactive Sonification: A Scoping Review

### Industry Projects

Human Interface Systems Lab — *Sep 2021 - Jan 2023*

- Subjective evaluation on in-vehicle virtual sound, *Harman International* (*Samsung Electronics*)
- Auditory display usability test for electric vehicles, *Hyundai MOBIS*
- Usability evaluation tool for virtual engine sound, *Hyundai Motor*
- Galvanic vestibular stimulation device development for pilots, *Republic of Korea Air Force*

### Undergraduate Projects

SNU and Korean Provincial Government — *Sep 2016 - Dec 2016*

- An ethnographic study on cheering culture of Korea University, Dr. Hilary Finchum-Sung
- An interpretation of clarinet repertoire from a performer's perspective, Dr. Eungi Min
- Silk Road cultural project across 10 ports in 9 countries as a musician, Gyeongsangbuk-do

### Clarinet Performance

Composition Presentations and Concert Performances — *2013 - Current*

- ANU Orchestra Concert, Australian National University, 2024
- Paris - Unwise Prince, Sungshin Women's University, 2022
- Reflection for 3 Clarinets in B flat, Ewha Womans University, 2017
- The Last Resort, Hanyang University, 2016
- Chamber Music Night, Ewon Art Hall, 2013
- Root Woodwind Quintet Recital, Yonsei University, 2013

## Achievements

2023-26 **PhD Scholarship**, Stipend for 3.5 years, travel and removal allowances — *ANU*

2023-26 **Fee Remission Merit Scholarship**, Doctoral program full tuition fee waiver — *ANU*

2021-22 **Scholarship for Next Academic Generation**, Master's program full tuition fee waiver — *SNU*

2017 **Undergraduate Independent Research Award**, Research grant and Encouragement award — *SNU*